

Wind Vanes in Ancient Mesopotamia, About 2000–1500 B.C.

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Abstract

A search made of the Sumerian and Akkadian literature for indications of the possible existence of wind vanes in the ancient Mesopotamian civilizations resulted in two discoveries. First, in one Akkadian fable, originally written between about 1800 and 1600 B.C., mention was made of a wind vane. It follows from the context of the fable that the vane was made of wood, while the name of the vane suggests that it was in the shape of a bird. Second, three Sumero-Akkadian vocabularies of this period give three different Sumerian names for the single Akkadian name for vane. The Sumerian names appear to be genuine Sumerian terms and not translations of the Akkadian term. All three Sumerian names suggest that the vanes were made of wood; one of the three may possibly indicate that the vane was made in the form of either a fish (shark?) or a mythological water monster. Since the Sumerian culture flourished before about 2000 B.C. (the Ur III dynasty ruled from about 2100 to 2000 B.C.), it seems that in the ancient Mesopotamian civilizations there were wind vanes about 4000 years ago, i.e., about 2000 years earlier than reported in ancient China and classical Greece.

1. Introduction

Hellmann (1908, p. 485) points out that the oldest known meteorological instrument is the wind vane. Literature quoted by Needham and Ling (1959, p. 478) indicates that wind vanes in the form of a thread or streamers already were in use in China in the second century B.C. Additionally, Needham and Ling cite a bronze wind gnomon in the shape of a bird in China in 101 B.C. As far as classical Greece is concerned, Vitruvius (1962, p. 57), the Roman architect and military engineer (flourished about 25 B.C.), tells us that Andronicus of Cyrrha's Tower of the Winds in Athens had a bronze wind vane in the shape of a triton at its top. According to Judeich (1931, p. 375), the Tower (and, presumably, the vane) dates back to the first century B.C. Thus, both the ancient Chinese and Greeks had wind vanes about 2000 years ago.

In view of the fact that the ancient Mesopotamian civilizations took a particularly active interest in the winds of their region, used the wind directions as geographical directions,

placed their buildings according to winds, etc. (see Neumann, 1977); it seemed worthwhile to make a search of the Sumerian and Akkadian literature for possible indications of the existence and use of wind vanes in these civilizations. This search resulted in the following: 1) A wind vane was mentioned in a fable that in all probability originally was composed in the Old Babylonian Period (OBP); that is, between about 1800 and 1600 B.C.; 2) Three Sumero-Akkadian lexical series (a kind of dictionary) have three different terms corresponding to the single Akkadian term for wind vane. These series also date back to OBP, and make it appear that the Sumerians had wind vanes at that time. This would mean that wind vanes were in use in Mesopotamia even before 2000 B.C.

Fig. 1 is a map of the Near East early in the second millennium B.C., and shows the locations of Sumer and Babylonia, etc.

2. Wind vane in a Babylonian Fable

One of the literary classes of composition cultivated in Babylonia during OBP was that of a special type of fable, the "contest fable." In these contest fables, two or more personified animals, plants, and even deities were made to debate the merits and value of their respective contributions. The fable of particular interest in this note is the "Fable of the Willow" (see, e.g., Lambert, 1960, pp. 164–167 and pl. 44), in which a willow² and other trees discuss the value of artifacts and tools made of them. Copies of the cuneiform text, a transliteration of the text into Latin characters, and its English translation are given by Lambert. The tablets are damaged, but fortunately most of the line mentioning the wind vane is legible. The tablet of special interest bears the museum number K 8413 in the Kuyunjik Collection of the British Museum. Lines 12–14 of the translation on p. 167 in Lambert's volume read as follows:

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² Akkadian *šarbatu*. This tree, identified as a willow by Lambert, is now generally identified as the Euphrates poplar (*Populus Euphratica*) (see discussion in the *Assyrian Dictionary of the Oriental Institute of the University of Chicago*, Vol. 16 (S) (Oppenheim *et al.*, 1962, p. 109)).

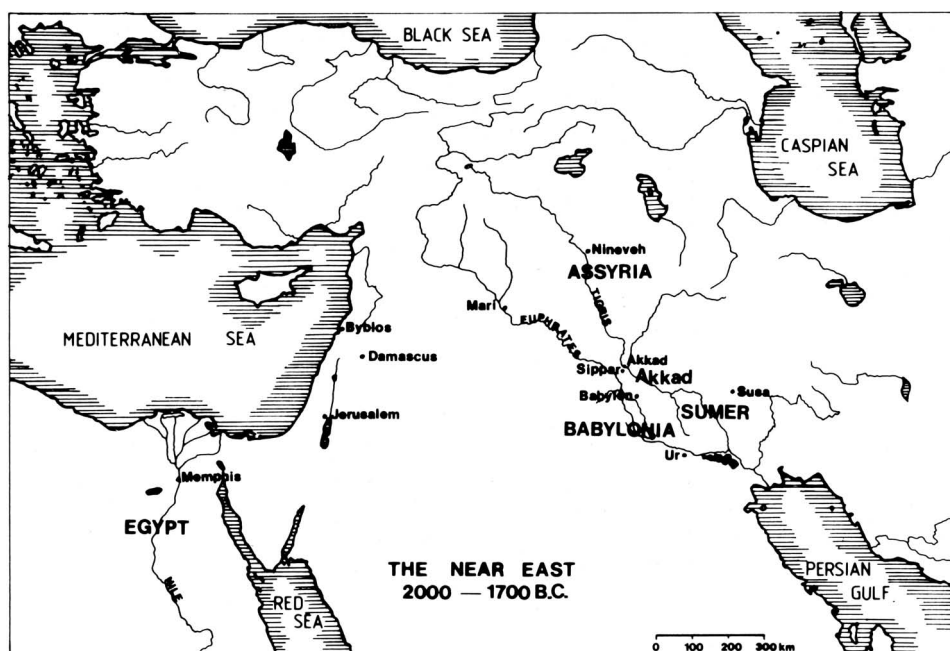


FIG. 1. Map of the Near East, 2000–1700 B.C., showing the location of Sumer, Babylon, etc. (actually Sumer ceased to exist by about 2000 B.C.) The areas of the former Sumer, Babylonia, and Assyria now are part of the present-day Iraq.

- 12 Like a crown the temple is adorned with . . .
 13 They look at the weather vane for the direction of
 the wind . . .
 14 . . . the sailor carries . . .

Line 13 makes it clear that the Babylonians possessed wind vanes at the time of the fable's composition. It also is interesting to speculate if line 12 means that the vane was on the roof of the temple (as is the case with Christian churches, at least from the Middle Ages on). And since the wind device is mentioned in a debate among trees, we may infer that the vane was made of wood.

The shape of the wind vane in OBP can be conjectured from its Akkadian name. In line 13 of the tablet just quoted (Lambert, 1960, p. 166), the pertinent term reads, in transcription, *iššūr šāri*. The literal meaning of this compound expression, "wind bird," (or more exactly, "bird [*iššuru*] of wind [*šāru*"]) suggests that the OBP wind vane normally was in the shape of a bird. (We recall that the Chinese of 100 B.C. also had their bronze vane in the shape of a bird. And the wind vane, or weather vane, of Christian churches has been in the shape of a cock for many centuries.)

Tablet K 8413 was written in about 650 B.C. for the library of the Assyrian king Assurbanipal (668–627 B.C.), who collected a large library in Nineveh, his capital. However, this tablet was only a copy for the royal library (Assurbanipal was in a way a "renaissance king," who collected literature) and it is almost certain that the original was written in OBP. Some indication of this is in the form ("genre") of the composition, but even more persuasive is the evidence arising from a consideration of the author's name. Lambert (1962) studied tablets in the British Museum that list the authorship of a number of cuneiform works. On p. 67 of his paper he gives a translation of a group of such tablets (his Group VI),

and in line 14 of the translation we read the statement:

- 14 *The Series of the Willow* [?]; by Urnanna, the
 magician, scholar of Babylon. [3]

Urnanna's name also is transcribed as Ur-Nanna, bringing out the elements of that composite name: "Ur" = servant (literally, "dog") and "Nanna" = the Sumerian moon deity. Composite names of this type became obsolete after ca. 1500 B.C., but still were very common in Babylon in OBP, as pointed out by Lambert (1962, p. 76, footnote 16):

If . . . Ur-Nanna is correctly located in Babylon in VI 14, he is almost certainly to be assigned to the Old Babylonian period. Names of the type Ur + nomen dei die out with the Old Babylonian period apart from a few archaistic examples in the Cassite period [1500–1200 B.C.], and during that period it is doubtful if a public figure in Babylon would bear a name compounded with the name of the moon god. However, in the Old Babylonian period it is very natural . . .

We are thus led to believe that the original willow fable was composed in OBP, and that therefore wind vanes must have been in use in Babylonia in about 1800–1600 B.C.

3. Sumero-Akkadian vocabularies

The Akkadian term for wind vane in the fable of the willow is, as was pointed out earlier, *iššūr šāri*. The entry for this

³ As to the question mark after "Willow," see footnote 2.

term in the *Assyrian Dictionary of the Oriental Institute of the University of Chicago* (CAD, for short) quotes (Oppenheim and Reiner, 1960, Vol. 7(I/J), p. 209a) three Sumerian equivalents of this term, located in three Sumero-Akkadian lexical series.

In the Harra-*hubullu* series we have three Sumerian names for wind vane: *giš.úh* (or *kušû*, see below), *giš.im.šēš*, and *giš.im.á.lá*. The second of these means “device to watch the wind”; the third could mean “device catching, or hanging, in the wind”. The first will be discussed later.

The second lexical series, giving Sumerian values of compound cuneiform symbols (‘graphemes’), is called *diri DIR siāku*, or *Diri* for short, and the appropriate entry is *giš-ku-ú-a GIŠ.UD+NAGAR+DIŠ = (GIŠ.KUŠŪ) = iš-sur ša-a-ri*.

The third lexical series, a bilingual version of *Diri*, is *ea A*, or *Ea* for short, and the relevant entry reads *ku-ia KUŠŪ = ša GIŠ.KUŠŪ iš-sur šāri*. The symbol (grapheme) *KUŠŪ* (with the Sumerian pronunciation *kūya*) stands for some aquatic animal or monster (Oppenheim and Reiner, 1971, CAD, Vol. 8(K), p. 602 under *kušû*, and Landsberger, 1962, p. 89 seqq.) and the suggestion offers itself that the pertinent Sumerian wind vane may have been in the form of a fish (perhaps shark) or a mythological water monster. This monster also is referred to by the first vane name in Harra-*hubullu* (*GIŠ.ÚH*), which actually is only a variant (‘graphemic’ variant) of *GIŠ.KUŠŪ*.⁴

As we see from these Sumerian names for wind vane, they all contain as the first element the “determinative” *gis*, which in Sumerian means wood. We are thus led to infer that the wind vanes of the Sumerians were in all probability made of wood, as was the case in OBP following the collapse of the Sumerian Ur III empire.

All three Sumerian terms for the Akkadian *iš-sur šāri* appear to be genuine Sumerian words that were probably taken over from third millennium usage or from lexical tradition. They are not simple translations of the Akkadian term; if this were the case, the translations would have read *im.mušen* or *mušen im.a* in Sumerian.

4. Summary

An Akkadian literary composition and three Sumero-Akkadian dictionaries dating back to OBP, or the period 1800–1600 B.C. in Mesopotamian history, indicate that wind vanes were

in existence and in use at that time. Since the Sumerian culture flourished before about 2000 B.C., the wind vanes in Mesopotamia must be over 4000 years old. The vanes were made of wood and apparently were in the form of birds or aquatic animals.

Until recently, the oldest known wind vanes, namely those of the Chinese and Greeks, dated back to the first and second centuries B.C., i.e., to a period about 2000 years ago. The present note raises the age of wind vanes by an additional 2000 years.

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⁴ *ÚH* is composed of the elements *UD*, *NAGAR*, and *DIŠ*, and also has the value *KUŠŪ*; correspondingly, *KUŠŪ* composed of the elements *NAGAR* and *DIŠ* can also be read *ÚH*.